
CF000: PRIMITIVE DISTINGUISHABILITY AND ORTHOGONAL ARTICULATION

A PREPRINT

Justin K. Lietz 
Neuroca, Inc.
justin@neuroca.ai
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ABSTRACT

This paper studies a deliberately early question: how can distinguishability begin before geometry, locality, carrier structure, calculus, or later physical machinery have been introduced? The answer developed here is a root formalism in which isolated nullity and isolated undifferentiated totality are not viable realized origins for a law-bearing branch. The admissible origin is instead a single primitive condition carrying unresolved pole-opposition internally. From that condition the paper derives non-flat distinguishability, the first admitted axis of articulation, same-axis saturation, and the forced move into a second irreducible axis. The result is a compact 0D–1D–2D formal spine for CF000: invariant-bearing articulation appears first, continuous same-axis bifurcation supplies the 1D effective invariant, and the second axis supplies the first numerical relation-grid in which algebra becomes meaningful. The paper is intended to be read as a constructive starting point for the later carrier and geometric stages, not as a finished theory of those later stages.

Keywords primitive origin, distinguishability, invariant-bearing articulation, orthogonal re-articulation, same-axis saturation, algebra unlock

Contents

Reader Orientation	4
Contribution and Scope	4
Companion Materials	4
How to Read the Formalism	4
1. Scope, Ontology, and Primitive Commitments	5
1.1 Root question	5
1.2 Declared closure and stage boundary	5
1.3 Primitive ontology	5
1.4 Stage boundaries and deferred structure	5
1.5 Status convention	5

2. Mathematical Setting and Definitions	6
2.1 Logical domain of discourse	6
2.2 Definitions	6
2.3 Primitive root laws	7
2.4 One-origin and no-prior-apartness discipline	7
2.5 Derived shorthand for the invariant	7
2.6 Finite proof language for structural witnesses	8
3. Foundational Construction: Forced Origin Architecture	8
3.1 Branch-scope witness	8
3.2 Nullity exclusion	8
3.3 Undifferentiated totality exclusion	8
3.4 Shared sterility does not collapse the poles	8
3.5 Exhaustive terminal classification at the primitive layer	9
3.6 Forced admissible origin theorem	9
3.7 Invariant-bearing precludes flatness	9
3.8 First non-flat articulation follows directly from the forced origin	9
3.9 Section summary	10
4. Main Theorems and Proofs	10
4.1 Dependence-ranked articulation chain	10
4.2 HRDCL - character of a genuinely higher-resolution degree	10
4.3 Invariant-bearing is constitutive of admitted articulation	11
4.4 HRDAP - admission of a genuinely higher-resolution degree	11
4.5 Contextual noncoincidence and apartness	11
4.6 ORS - pre-metric orthogonal rotation structure	11
4.7 Same-axis saturation and forced orthogonal re-articulation	12
4.8 0D -> 1D -> 2D closure	12
4.8A Axis-proper effective invariants at first and second admission	13
4.9 Algebra unlock	14
4.10 Recursive restatement of the invariant	14
4.11 Section summary	14
5. Derived Physical Consequences	15
5.1 A8 visibility note	15
5.2 Forward note on effective invariants	15
5.3 A forward empirical question	15
5.4 Interpretive extensions	15
6. Structural Validation by Finite Witnesses	15
6.1 Exact finite checks	15

6.2 Primitive-candidate enumeration witness	16
6.3 Counterexample attempts	16
6.4 Saturation-band decision witness	16
6.5 Relation-grid exhaustive scan	17
7. Worked Example or Minimal Realization	18
7.1 Minimal realization setup	18
7.2 Expected behavior	18
7.3 What the example should show	18
7.4 Algebraic witnesses of the invariant spine	18
8. Companion Notebook and Executable Traceability	19
8.1 Notebook traceability map	20
9. Assumptions, Scope, and Open Questions	20
9.1 Assumptions used in the derivation	20
9.2 Scope of applicability	20
9.3 Claims established here	20
9.4 Claims left for later work	21
9.5 Open questions and later handoff	21
10. Conclusion	21
11. Integration with Broader VDM Theory	21
12. References and Provenance	21
12.1 Internal constitutional / provenance and downstream alignment references	22
12.2 Structural and executable companion artifacts	22
12.3 Local symbol and closure summary	22

Reader Orientation

This paper begins at the pre-carrier layer of the present VDM branch. Its purpose is to make the first step of distinguishability explicit before later structures such as geometry, locality, differentiability, QGT, or metriplectic dynamics are brought into the story.

The central object is one primitive origin-condition. It is not a point in a background space and it is not a pair of already separated things. It is a single admissible condition in which the two terminal poles, absolute nullity and absolute undifferentiated totality, remain unresolved while being borne internally. The constructive question is then simple: once that unresolved burden is admitted, what must be carried next?

The answer developed below is the 0D–1D–2D spine. The origin first yields non-flat articulation; the first admitted axis carries continuous same-axis bifurcation; saturation of that axis forces a second irreducible axis; and only at that two-axis stage does the first numerical relation-grid, and therefore algebraic description, become available.

Contribution and Scope

The paper contributes a self-contained root formalism for the earliest admissible route from primitive unresolved origin to two-axis articulation. The main contribution is not a numerical simulation and not a later physical model. It is a dependency-clean derivation of the first articulation structure used by the downstream CF stack.

The work establishes:

1. the exclusion of isolated nullity and isolated undifferentiated totality as realized branch roots,
2. the forced admissible origin as one condition bearing unresolved primitive opposition,
3. the first non-flat invariant-bearing articulation,
4. same-axis saturation and forced orthogonal re-articulation,
5. 0D–1D–2D closure,
6. the 1D effective invariant of continuous same-axis bifurcation,
7. the 2D numerical relation-grid invariant and the first algebraic unlock.

Later structures are kept in their proper place. Carrier construction, QGT, metric curvature, Laplacian structure, differentiability, and the named J/M machinery belong downstream. They are valuable continuations of the program, but they are not used as primitive inputs here.

Companion Materials

A companion notebook is included to make the paper easier to explore computationally. The notebook does not replace the derivation. It gives small executable witnesses for the paper’s main structural claims: primitive candidate comparison, the 0D–1D–2D stage map, same-axis saturation, and the first two axis-proper effective invariants.

The release also includes a Lean 4 surface file and a SymPy script for the finite witness layer. Their role is modest and useful: they make the primitive-survivor count, saturation index, and relation-grid counts easy to inspect as exact Boolean and integer claims.

The notebook is written as a readable exploration. Its role is to help a reader see the formal spine move in a concrete finite witness model, collect the key measurements, and provide a reproducible starting point for later CFN work.

How to Read the Formalism

The manuscript uses a strict dependency order. Definitions appear before the theorems that use them; later machinery is mentioned only as downstream context; and empirical consistency notes are separated from theorem-bearing claims.

Status labels such as “Proved in this CF,” “Derived structural consequence,” and “Forward note” are used to help the reader know what kind of claim is being made. They are not meant to turn the paper into a compliance report. They are a compact guide to where the mathematical burden lives and where later work naturally continues.

The short version is: Sections 1–4 carry the root derivation, Sections 5–7 show consequences and minimal witnesses, Section 8 explains the notebook pairing, and Sections 9–11 collect scope, integration, provenance, and reusable notation.

1. Scope, Ontology, and Primitive Commitments

1.1 Root question

CF00 begins later than the absolute root. CF000 exists because that is already too late for an origin account. The root question of this document is: What is the earliest admissible primitive law from which distinguishability, higher articulation, and the first lawful orthogonal re-articulation can be earned without primitive geometry, carrier structure, locality, time, causality, valuation, parameterization, or differentiability?

1.2 Declared closure and stage boundary

This CF closes only the staged route:

0D -> 1D -> 2D. Here that means:

- 0D: primitive unresolved origin before admitted axis-specific articulation,
- 1D: first admitted axis of non-flat invariant-bearing articulation,
- 2D: the first orthogonal re-articulation forced when the 1D axis is exhausted without discharge. At each admitted axis/domain within this closure, the invariant is not merely preserved but re-expressed as a new axis-proper effective invariant proper to that admission: first at 1D as continuous same-axis bifurcation and again at 2D as the numerical relation-grid invariant. This CF does not prove QGT, metric geometry, carrier-family mechanics, differentiability, or the named conservative/dissipative split. Those belong later.

1.3 Primitive ontology

The primitive vocabulary is deliberately small. CF000 starts with the terminal pole marks, their non-identity, their sterility for realized differentiated articulation, their opposition, an origin-candidate predicate, a non-vacuousness predicate, a realizability predicate, a flatness predicate, and the one-origin invariant discipline.

From that compact vocabulary, the paper derives the following structures: primitive invariant carriage, differentiated articulation, the lowest-resolution invariant-bearing degree, genuinely higher-resolution degree, recursive restatement, contextual noncoincidence, apartness, pre-metric orthogonal re-articulation, same-axis saturation, the 1D axis-proper effective invariant, continuous same-axis bifurcation, serial/proto-natural-number structure at 1D, inherited invariant stacking at 2D, the 2D numerical relation-grid invariant, 2D closure, and algebra unlock.

1.4 Stage boundaries and deferred structure

The following structures are intentionally deferred at the CF000 layer. They are not rejected; they are simply not primitive resources for this paper's root derivation:

- carrier language,
- topology, metric structure, or chart language,
- locality or neighborhood structure,
- time parameter, event sequence, propagation, or causality,
- graph, node, edge, or tree language,
- QGT,
- named J, M, or $J \oplus M$ machinery,
- differentiability, Laplacian, or smooth support structure.

Keeping these objects downstream is what lets CF000 state its own burden cleanly: the first articulation has to be earned before the later carrier and geometric vocabulary can be used.

1.5 Status convention

Every major claim in this document is labeled as one of:

- Proved in this CF

- Derived structural consequence
- Structural witness note
- Forward note for CF00
- Interpretive / out-of-scope extension

2. Mathematical Setting and Definitions

2.1 Logical domain of discourse

Let $\mathcal{C}$ denote the logical domain of candidate realization conditions. This is not a primitive ontology of objects in a world. It is only a logical range of discourse, introduced so that realizability claims can be stated without presupposing points, carriers, graphs, or manifolds.

2.2 Definitions

Absolute nullity. Complete realized absence.

Formal symbol: distinguished pole mark **0**.

Status: *Primitive*.

Absolute undifferentiated totality. Complete homogeneous one-ness without internal noncoincidence.

Formal symbol: distinguished pole mark **1**.

Status: *Primitive*.

Origin candidate. Candidate primitive condition for a realized branch.

Formal symbol: predicate $\text{Orig}(x)$ on $\mathcal{C}$.

Status: *Primitive*.

Sterile. Cannot admit realized differentiated articulation while remaining what it is.

Formal symbol: predicate $\text{Ster}(x)$.

Status: *Primitive*.

Flat. Contains no realized internal noncoincidence by which differentiated articulation could be sustained.

Formal symbol: predicate $\text{Flat}(x)$.

Status: *Primitive*.

Non-vacuous. Not complete realized absence in the root sense.

Formal symbol: predicate $\text{NV}(x)$.

Status: *Primitive*.

Realizable. Admissible as a non-vacuous realized branch-condition.

Formal symbol: predicate $\text{Real}(x)$.

Status: *Primitive*.

Opposition. Non-identity of the two terminal poles together with their shared sterility verdict.

Formal symbol: predicate $\text{Opp}(x, y)$.

Status: *Primitive*.

Primitive invariant. The one admissible primitive origin-condition bears unresolved pole-opposition internally and does not terminate in pole-discharge.

Formal shorthand: $\text{Inv}(x)$.

Status: *Derived shorthand*.

Differentiated articulation. A realizable condition that is not flat:

$$\text{Diff}(x) := \text{Real}(x) \wedge \neg \text{Flat}(x).$$

Status: *Derived*.

Lowest-resolution invariant-bearing differentiated degree. The invariant occurs in realizable non-flat articulation at the least admitted degree after origin:

$$\text{Deg}_{\min}(x) := \text{Inv}(x) \wedge \text{Diff}(x).$$

Status: *Derived*.

Genuinely higher-resolution differentiated degree. A higher articulation of the same invariant that adds an internal determination rather than merely renaming the lower one.

Status: Derived target notion.

Pre-metric orthogonal re-articulation. Admission of a new axis/domain irreducible to the exhausted current axis/domain, without yet assuming metric orthogonality.

Status: Derived target notion.

Same-axis saturation. The current admitted axis/domain is same-axis saturated when no genuinely new lawful invariant-bearing articulation remains available within that same axis/domain. Any further purported articulation there would either be mere renaming or reparameterization of what is already borne, or would require a new irreducible axis/domain and therefore not belong to the same axis/domain. Same-axis saturation is not discharge, not resolution of the invariant, not collapse, not termination, and not a claim that nothing further happens. It is only exhaustion of lawful same-axis expression.

Status: Derived target notion.

Algebra unlock. The first licensing of ordered-pair description and lawful binary combination once two irreducible axes are admitted.

Status: Derived consequence.

2.3 Primitive root laws

The present draft adopts the following primitive root laws.

PB1. Pole non-identity

$$0 \neq 1$$

PB2. Pole sterility

$$\text{Ster}(0) \wedge \text{Ster}(1)$$

PB3. Pole opposition

$$\text{Opp}(0, 1)$$

PB4. Realizability implies non-vacuousness and pole exclusion

$$\forall x \in C,$$

$$\text{Real}(x) \Rightarrow \text{NV}(x) \wedge x \neq 0 \wedge x \neq 1.$$

2.4 One-origin and no-prior-apartness discipline

The manuscript adopts the following root discipline.

- There is initially no prior basis under which the two poles are primitive separate things.
- Therefore the poles may not be read as two externally separated primitive things.
- Any admissible primitive origin must be read as one primitive condition bearing the opposition internally. This is a root discipline of the rewrite. It is not permission to smuggle in an unearned mediating substance.

2.5 Derived shorthand for the invariant

When convenient, write $\text{Inv}(x)$ for the following schematic situation:

- x is an admissible primitive origin-condition,
- x does not coincide with either isolated pole,
- the non-identical sterile poles are opposed,
- that opposition is internal to the one origin-condition,
- and the condition does not terminate in pole-discharge. This shorthand is mnemonic. It does not replace proof burden.

2.6 Finite proof language for structural witnesses

The proofs above are stated in prose because the root vocabulary is intentionally prior to geometry and carrier machinery. For the finite structural witnesses in Section 6, the same burden is written in a small Boolean language so the reader can check the witness tables exactly.

For a candidate x , define the finite admissibility test

$$\text{Adm}(x) := \text{NV}(x) \wedge \neg \text{Flat}(x) \wedge \text{BearOpp}(x) \wedge \text{OneOrigin}(x) \wedge \text{NoDischarge}(x).$$

Here $\text{BearOpp}(x)$ records that the primitive pole-opposition is borne internally, $\text{OneOrigin}(x)$ records that the poles have not been externalized into two prior primitive things, and $\text{NoDischarge}(x)$ records that the branch has not collapsed into either isolated terminal pole. In the witness tables, each predicate is evaluated as 0 or 1 and a candidate survives exactly when $\text{Adm}(x) = 1$.

The same finite language is also used for the saturation and relation-grid witnesses:

$$S(n) = \min(n, 9), \quad G_2 = \{0, 1, 2, 3\} \times \{0, 1, 2, 3\}, \quad R_2 = G_2 \setminus \{(0, 0)\}.$$

The first expression is the declared same-axis decision witness used in Section 6.4: after index 9, a further genuinely new burden is not counted as same-axis novelty. The second and third expressions define the smallest relation-grid witness used in Section 6.5: the 2D grid has 16 placements and 15 nonzero relation vectors from the origin. These are exact finite-cardinality checks, so the numerical tolerance is 0.

3. Foundational Construction: Forced Origin Architecture

3.1 Branch-scope witness

Definition 3.1.1.

An origin candidate is any $x \in C$ such that $\text{Orig}(x)$ holds.

Scope Convention 3.1.2 (The branch under study is realized).

This manuscript classifies the primitive origin of a realized falsifiable branch. Let ω denote the primitive origin-condition of the branch under study. Then

$$\text{Orig}(\omega) \wedge \text{Real}(\omega).$$

This is not a detachable extra premise about arbitrary worlds. It is the witness fixed by the subject matter of the manuscript itself.

3.2 Nullity exclusion

Theorem 3.2.1 (Nullity exclusion).

Absolute nullity is not a viable realized root for a falsifiable law-bearing branch. Status: Proved in this CF. Proof. A falsifiable law-bearing branch must support at least the distinction between admissible and inadmissible articulation and between realized and unrealized burden. Absolute nullity supports no realized content and no realized distinction. Therefore it cannot serve as the realized root of the present branch. $\square$

3.3 Undifferentiated totality exclusion

Theorem 3.3.1 (Undifferentiated totality exclusion).

Absolute undifferentiated totality is not a viable realized root for a falsifiable law-bearing branch. Status: Proved in this CF. Proof. A falsifiable law-bearing branch requires some realized noncoincidence by which one admissible condition can fail to coincide with another or with itself in a relevant way. Absolute undifferentiated totality contains no internal noncoincidence and therefore cannot sustain differentiated realized articulation. Hence it cannot serve as the realized root of the present branch. $\square$

3.4 Shared sterility does not collapse the poles

Proposition 3.4.1.

The poles 0 and 1 are not identical even though both are sterile for the present branch. Status: Proved in this CF. Proof. By PB1, $0 \neq 1$. By PB2, both are sterile. Distinctness and shared sterility therefore coexist. $\square$

3.5 Exhaustive terminal classification at the primitive layer

Definition 3.5.1 (Terminal primitive pole-candidate).

A terminal primitive pole-candidate is a primitive origin-candidate considered only at terminal, flat, sterile resolution: no internal noncoincidence is available by which differentiated articulation could be admitted, and no added basis is available by which the candidate could be decomposed or separated.

Theorem 3.5.2 (No third terminal primitive pole survives).

Let t be any terminal primitive pole-candidate. Then exactly one of the following holds: 1. $t = 0$, 2. $t = 1$. In particular, no third terminal primitive pole survives. Status: Proved in this CF. Proof. At the primitive layer, classification can use only the weakest distinctions already on the page: whether any realized content is present at all, and whether any internal noncoincidence is available.

- If no realized content is present, the candidate is complete realized absence and so equals 0.
- If realized content is present, the candidate is non-vacuous; but if it is also terminal, flat, and primitive, then no internal noncoincidence is available by which differentiated articulation could be admitted, and no richer content may be imported from outside. Therefore realized content is present yet internally undifferentiated, so the candidate collapses to 1. There is no third case. Any purported third primitive terminal pole either lacks realized content, possesses realized content without internal noncoincidence, or already contains richer articulated burden and therefore is not a terminal primitive pole-candidate. Hence no third terminal primitive pole survives. $\square$

3.6 Forced admissible origin theorem

Theorem 3.6.1 (Forced admissible origin theorem).

For the realized branch under study, the primitive origin-condition ω is one primitive condition bearing the opposition of 0 and 1 internally without terminal settlement. Equivalently,

$\text{Inv}(\omega)$. Status: Proved in this CF. Proof. By Scope Convention 3.1.2, ω is the primitive origin-condition of a realized falsifiable branch. By Theorem 3.2.1, $\omega \neq 0$. By Theorem 3.3.1, $\omega \neq 1$. By Theorem 3.5.2, there is no third terminal primitive pole. Therefore the realized branch-origin cannot be any terminally settled primitive pole at all. By Section 2.4, the primitive layer contains no prior basis under which two primitive poles could begin as externally separate things. So the relevant primitive opposition cannot be realized as two already-distinct primitive objects. By Proposition 3.4.1, the two poles are genuinely distinct even though both are sterile in isolation. Since the realized branch-origin cannot terminate in either isolated pole, cannot be a third terminal primitive pole, and cannot externalize the poles into two already-separated primitive things, the only admissible primitive possibility left is that the distinction between the two terminal poles is borne internally by one primitive origin-condition without terminal settlement. That is exactly the manuscript meaning of $\text{Inv}(\omega)$. $\square$

Corollary 3.6.2 (No simpler primitive survivor remains).

Relative to the inferential discipline of this manuscript, no simpler primitive survivor remains beyond: 1. isolated nullity, 2. isolated undifferentiated totality, 3. one primitive origin-condition bearing their unresolved internal opposition. The first two are excluded for any realized falsifiable branch, so only the third remains admissible. Status: Proved in this CF.

3.7 Invariant-bearing precludes flatness

Theorem 3.7.1 (Invariant-bearing is incompatible with flatness).

For any $x \in C$,

$\text{Inv}(x) \Rightarrow \neg \text{Flat}(x)$. Status: Proved in this CF. Proof. Flatness means that no realized internal noncoincidence is available by which differentiated articulation could be sustained. But $\text{Inv}(x)$ states precisely that unresolved pole-opposition is internally borne by one admissible origincondition without terminal discharge. A condition bearing unresolved internal opposition is therefore not flat. $\square$

3.8 First non-flat articulation follows directly from the forced origin

Theorem 3.8.1 (Breakdown of perfect flatness at the realized origin).

For the realized branch under study,

$\neg \text{Flat}(\omega)$. Status: Proved in this CF. Proof. By Theorem 3.6.1, $\text{Inv}(\omega)$. Apply Theorem 3.7.1. $\square$

Corollary 3.8.2 (Differentiated articulation is present at the realized origin).

For the realized branch under study,

$\text{Diff}(\omega)$. Status: Proved in this CF.

Proof.

By Scope Convention 3.1.2, $\text{Real}(\omega)$. By Theorem 3.8.1, $\neg \text{Flat}(\omega)$. Apply the definition of Diff . $\square$

Corollary 3.8.3 (Lowest-resolution invariant-bearing differentiated degree is witnessed).

In witness form, the present branch satisfies

$\exists x \in C \text{ Orig}(x) \wedge \text{Degmin}(x)$,

with witness $x = \omega$. Status: Proved in this CF. Proof.

By Theorem 3.6.1 and Corollary 3.8.2, $\text{Inv}(\omega) \wedge \text{Diff}(\omega)$, hence $\text{Degmin}(\omega)$. $\square$

3.9 Section summary

At this stage the manuscript has:

- proved exclusion of isolated nullity,
- proved exclusion of isolated undifferentiated totality,
- proved distinctness of the poles despite shared sterility,
- proved exhaustive terminal classification at the primitive layer,
- proved that no third terminal primitive pole survives,
- proved that the only admissible primitive origin left is one origin-condition bearing unresolved pole-opposition,
- proved that no simpler primitive survivor remains for a realized falsifiable branch,
- proved incompatibility of invariant-bearing with flatness,
- proved differentiated articulation at the realized origin,
- and proved the lowest-resolution invariant-bearing differentiated degree.

4. Main Theorems and Proofs

4.1 Dependence-ranked articulation chain

The articulation chain must now be read in dependence-ranked form: 1. one admissible origin-condition carries unresolved internal pole-opposition, 2. a lowest-resolution invariant-bearing differentiated degree is admitted, 3. the question becomes whether that degree admits a genuinely higher-resolution differentiated degree under the same invariant, 4. if such a degree is admitted, the lower degree is no longer sufficient as sole determination, 5. within an admitted axis/domain, every genuinely new lawful same-axis invariant-bearing articulation belongs to the articulation burden of that axis/domain, 6. once no genuinely new lawful same-axis articulation remains, same-axis saturation holds,

7. if same-axis saturation obtains while discharge remains impossible, orthogonal re-articulation is forced, 8. the first such forced orthogonal re-articulation closes the move into 2D, 9. only then is algebra licensed.

This is not chronology. It is dependency order.

4.2 HRDCL - character of a genuinely higher-resolution degree

Theorem 4.2.1 (Higher-Resolution Degree Character Lemma).

If a lowest-resolution invariant-bearing differentiated degree admits a genuinely higher-resolution differentiated degree under the same invariant, then:

1. the lower degree is no longer sufficient as the sole determination of the state, 2. an additional internal determination is present, 3. metric geometry, carrier structure, and differentiability still do not follow from this theorem alone.

Status: Proved in this CF. Proof. A genuinely higher-resolution differentiated degree is not a vacuous restatement of the lower one but a more articulated inhabitation of the same invariant. If the lower degree remained sufficient as sole determination, no added internal determination would have been introduced and the alleged higher degree would collapse into renaming. That contradicts genuineness. Hence the lower degree is no longer sufficient and an additional internal determination is present. But this alone does not state metric separation, carrier structure, or differentiability. $\square$

4.3 Invariant-bearing is constitutive of admitted articulation

Theorem 4.3.1 (Non-bearing is contradictory).

Let x be an admitted branch articulation. Then $\neg \text{Inv}(x)$ is contradictory. Status: Proved in this CF. Proof. Assume x is an admitted branch articulation and $\neg \text{Inv}(x)$. Then the primitive unresolved opposition is not borne in x . That is discharge. Terminal discharge resolves the branch-origin burden toward a terminal pole. By Theorem 3.5.2, no third terminal option survives; so discharge collapses toward either 0 or 1. But collapse toward 0 contradicts Theorem 3.2.1 and collapse toward 1 contradicts Theorem 3.3.1 for any realized falsifiable branch. Therefore non-bearing is contradictory. $\square$

Corollary 4.3.2.

Invariant-bearing is not an added persistence claim above admission. It is constitutive of admission. Status: Derived structural consequence.

4.4 HRDAP - admission of a genuinely higher-resolution degree

Theorem 4.4.1 (Higher-Resolution Degree Admission Principle).

At least one origin-bearing Degmin admits a genuinely higher-resolution differentiated degree under the undischarged primitive invariant. Status: Proved in this CF. Proof. Assume HRDAP fails. Then every origin-bearing Degmin is exhaustively complete as the terminal articulation of the invariant in that degree. Exhaustive completion is terminal settlement. Terminal settlement resolves the unresolved pole-opposition. By Theorem 3.5.2, no third terminal option survives. So terminal settlement collapses toward either 0 or 1, both excluded by Theorems 3.2.1 and 3.3.1 for any realized falsifiable branch. Contradiction. Therefore exhaustive completion of Degmin is false, and at least one origin-bearing Degmin admits a genuinely higher-resolution differentiated degree. $\square$

4.5 Contextual noncoincidence and apartness

Theorem 4.5.1 (Contextual Noncoincidence).

Two distinct higher-resolution determinations of the same invariant at the same articulation level are not identical as the same determination. Status: Proved in this CF. Proof. Suppose two allegedly distinct higher-resolution determinations $D1$ and $D2$ are identical. Then no new determination exists at that level. The level is exhaustively complete by a single determination. But by the contradiction used in

Theorem 4.4.1, exhaustive completion forces terminal settlement and collapse toward an excluded pole. Therefore $D1 \neq D2$. $\square$

Corollary 4.5.2 (Apartness is licensed).

Once contextual noncoincidence is proven between same-level determinations, primitive apartness of determination is licensed. Status: Proved in this CF. Proof. Apartness here means only that distinct same-level determinations are not the same determination. Theorem 4.5.1 gives exactly that. No metric structure is implied. $\square$

4.6 ORS - pre-metric orthogonal rotation structure

Theorem 4.6.1 (Orthogonal Rotation Structure, pre-metric form).

Let $D1$ and $D2$ be two contextually noncoincident higher-resolution determinations of the same invariant at the same articulation level. If the current axis/domain of articulation is exhaustively saturated by $D1$, then admissibility of

D2 requires a new axis/domain irreducible to the saturated one. This irreducible re-articulation is the first pre-metric orthogonal rotation.

Status: Proved in this CF. Proof. By Theorem 4.5.1, D1 and D2 are not identical. Suppose nevertheless that D2 is only further expression along the same exhaustively saturated axis/domain as D1. Then no genuinely new admissible determination is available there; the axis/domain is already complete. That is same-domain exhaustion. Since non-bearing is contradictory by Theorem 4.3.1, the invariant cannot terminate there. Therefore the admissibility of D2 cannot be realized as more of the same exhausted axis/domain. A new axis/domain, irreducible to the first, is required. This is exactly the manuscript meaning of pre-metric orthogonal re-articulation. $\square$

4.7 Same-axis saturation and forced orthogonal re-articulation

Definition 4.7.1 (Same-axis saturation).

The current admitted axis/domain is same-axis saturated when no genuinely new lawful invariant-bearing articulation remains available within that same axis/domain. Any further purported articulation there would either be mere renaming or reparameterization of what is already borne, or would require a new irreducible axis/domain and therefore not belong to the same axis/domain. Clarification. Same-axis saturation is not discharge, not resolution of the invariant, not collapse, not termination, and not a claim that nothing further happens. It is only exhaustion of lawful same-axis expression.

Theorem 4.7.2 (Exhaustive bearing within an admitted axis/domain).

Within a currently admitted axis/domain, any genuinely new lawful same-axis invariant-bearing articulation belongs to the articulation burden of that same axis/domain. It cannot be omitted and still count as lawful completion of that axis/domain. Status: Proved in this CF. Proof. Let the current axis/domain be admitted. Suppose a genuinely new lawful same-axis invariant-bearing articulation remains available there and yet is left outside the articulation burden of that same axis/domain. Then the axis/domain is being treated as lawfully complete while lawful same-axis invariant-bearing articulation still remains unborne there. That is not completion of the current axis/domain at all; it is incomplete same-axis bearing falsely treated as closure. Since Theorem 4.3.1 makes non-bearing contradictory for admitted articulation, lawful same-axis invariant-bearing articulation cannot be discarded as though it were irrelevant to the current axis/domain. Therefore any genuinely new lawful same-axis invariant-bearing articulation belongs to the articulation burden of that same axis/domain and cannot be omitted while claiming lawful completion there. $\square$

Theorem 4.7.3 (Saturation theorem).

Within a given admitted orthogonality class, once lawful same-axis expression of the invariant is exhaustively saturated, further admitted articulation in that same class is impossible without contradiction. Since terminal discharge is excluded, orthogonal re-articulation is forced. Status: Proved in this CF.

Proof. By Definition 4.7.1, same-axis saturation means no genuinely new lawful invariant-bearing articulation remains available within the current axis/domain. Suppose nevertheless that a further admitted articulation exists in that same class. If it is genuinely new and lawful, then Definition 4.7.1 was false and the axis/domain was not saturated after all. If it is not genuinely new, then it is only renaming or reparameterization of what is already borne and therefore not a new admitted articulation in that class. Contradiction. Therefore no further admitted articulation exists there once same-axis saturation holds. But if admitted articulation ceased altogether, the invariant would have no borne expression and would be discharged, contradicting Theorem 4.3.1. Hence further articulation must occur, but not in the saturated class. By

Theorem 4.6.1, it is forced to occur as orthogonal re-articulation. $\square$

4.8 0D $\rightarrow$ 1D $\rightarrow$ 2D closure

Definition 4.8.1 (Stage labels).

- 0D: the primitive origin stage, prior to admitted axis-specific articulation.
- 1D: the first admitted axis/domain of non-flat invariant-bearing articulation.
- 2D: the first stage at which a second irreducible axis/domain has been admitted by orthogonal re-articulation.

Theorem 4.8.2 (0D $\rightarrow$ 1D).

The forced origin together with Theorem 3.8 closes the move from 0D to 1D. Status: Proved in this CF. Proof. At 0D, the invariant is borne at the primitive origin without yet resolved axis-specific articulation. By Theorem 3.8, the realized origin is non-flat and differentiated. That is the first admitted axis/domain of articulated expression. Hence 0D

-> 1D is closed. $\square$

Theorem 4.8.3 (1D -> 2D).

The same-axis saturation theorem together with ORS closes the move from 1D to 2D. Status: Proved in this CF. Proof. By Theorem 4.7.2, lawful completion of 1D requires exhaustive same-axis bearing of genuinely new lawful same-axis articulation. By Definition 4.7.1, that is exactly same-axis saturation. Same-axis saturation is not discharge by the clarification in Section 4.7, and discharge is excluded by Theorem 4.3.1. Therefore lawful closure of 1D cannot terminate there. By Theorems 4.6.1 and 4.7.3, further articulation must occur as irreducible orthogonal re-articulation into a new class. The first such new class is exactly 2D. Hence 1D -> 2D is closed. $\square$

4.8A Axis-proper effective invariants at first and second admission

Definition 4.8.4 (Continuous same-axis bifurcation, pre-metric sense).

At the present stage, “continuous” means only that once a first axis/domain is admitted, lawful same-axis invariant-bearing articulation is indefinitely reiterable in that axis/domain until same-axis saturation is reached. Bifurcation here is not an additional postulate. It is the direct expression of the primitive two-pole opposition along the admitted axis/domain: each iterated articulation step inherits the irreducible polar structure of the invariant itself, presenting two irreconcilable directions of continuation. No metric continuum, coordinate parameter, calculus, or topology is implied.

Theorem 4.8.5 (Axis-proper effective invariant at 1D admission).

Once 1D is admitted, the invariant is not merely borne on an axis/domain as a passive container condition. The admitted 1D axis/domain immediately acquires a new effective invariant proper to that axis/domain: continuous same-axis bifurcative articulation under the inherited primitive invariant. Status: Proved in this CF.

Proof. By Theorem 4.8.2, 1D is the first admitted axis/domain of articulated invariant-bearing expression. By Theorem 4.3.1, invariant-bearing is constitutive of admission, so the admitted axis/domain cannot be neutral with respect to the invariant it bears. By Theorem 4.10.1, any realizable invariant-bearing articulation that remains non-exhaustive relative to the undischarged primitive invariant requires a further articulated condition again under that same invariant. Applied to first admission, lawful continuation in 1D is therefore not merely occupancy of a line but reiterated same-axis articulation under the same invariant. Since 1D is the first admitted axis/domain, this reiterated same-axis articulation is proper to that axis/domain itself. By Definition 4.8.4, this is continuous same-axis bifurcation. It is therefore a new effective invariant proper to 1D, borne together with the inherited primitive invariant. $\square$

Corollary 4.8.6 (1D bears the invariant twice).

At 1D, the branch bears the invariant in two non-identical but inseparable senses: first as inherited primitive unresolved burden, and second as the axis-proper effective invariant of continuous same-axis bifurcation. Status: Proved in this CF.

Proof. Immediate from Theorem 4.8.5 together with the continuing force of Theorem 4.3.1. $\square$

Corollary 4.8.7 (Seriality and proto-number at 1D).

Because the admitted 1D axis/domain supports continuous same-axis bifurcative articulation until saturation, 1D is the first stage at which seriality, repeatable more/less, and proto-natural-number structure become licensed. No later number-theoretic completion is claimed here. Status: Derived structural consequence.

Proof. Continuous same-axis bifurcative articulation licenses reiterable lawful distinction along one admitted axis/domain. That is sufficient for serial order, repeatable succession, and more/less structure. Because the invariant cannot discharge, articulation along 1D is non-terminating; each bifurcative step generates a successor without ceiling, which is exactly proto-natural-number structure. $\square$

Theorem 4.8.8 (Inherited stack and second-axis effective invariant at 2D admission).

When 2D is admitted, the prior invariant stack is inherited rather than erased: the primitive invariant and the 1D effective invariant remain borne, and the newly admitted second axis/domain immediately acquires its own effective invariant proper to that axis/domain. Status: Proved in this CF.

Proof. By Theorem 4.8.3, 2D is admitted only by orthogonal re-articulation from a same-axis-saturated 1D that still cannot discharge the invariant. Therefore admission of 2D cannot annihilate what 1D had already lawfully borne; otherwise discharge or loss of borne invariant-content would occur, contradicting Theorem 4.3.1. The prior invariant burden is therefore inherited. But by the same logic used in Theorem 4.8.5, a newly admitted axis/domain is not a neutral container: once admitted, it immediately enters lawful invariant-bearing articulation proper to itself. Hence 2D inherits the prior stack and adds a new second-axis effective invariant. $\square$

Corollary 4.8.9 (Numerical relation-grid at 2D).

Because 2D inherits 1D serial bifurcation and adds a second axis/domain with its own axis-proper effective invariant, the first grid of numerical relation becomes possible at 2D. This is the structural reason algebra first becomes admissible only there. Status: Derived structural consequence.

Proof. Two admitted irreducible axes carrying inherited and new bifurcative burden jointly license pair-placement and cross-axis relation. A cross-axis relation is a determination whose specification requires reference to both axes simultaneously and cannot be decomposed into a determination on either axis alone; that is exactly what the irreducibility of the two axes from each other first makes possible at 2D. That is exactly the first grid of numerical relation presupposed by Theorem 4.9.1. $\square$

4.9 Algebra unlock**Theorem 4.9.1 (Algebra unlock at 2D).**

Once 2D is admitted, ordered-pair description and lawful binary combination are licensed. This is the first algebraic unlock of the branch. Status: Proved in this CF.

Proof. By Theorem 4.8.3, two irreducible articulation axes/classes are admitted. With two irreducible axes, a determination may be specified by its two-axis placement rather than by a single-axis burden alone. This licenses ordered-pair description (a, b) relative to the admitted pair of axes. Once ordered pairs are licensed, lawful binary combination and comparison of coefficients on those axes are also licensed. That is the first algebraic regime of the branch. No richer algebraic structure is claimed here. $\square$

4.10 Recursive restatement of the invariant**Theorem 4.10.1 (Recursive Reapplication Principle).**

If a realizable invariant-bearing articulation is non-exhaustive relative to the undischarged primitive invariant, then a further articulated condition is required that again stands under that invariant. Status: Proved in this CF. Proof. The contradiction used to establish HRDAP does not depend on the specific articulation level being Degmin . The only resources used were: realized branch admissibility, non-discharge of the invariant, terminal-pole exclusion, and the no-third-terminal result. Therefore any admitted articulation that still leaves the invariant non-exhaustive cannot be terminal. A further articulation is then required under the same invariant. $\square$

4.11 Section summary

At the end of Section 4, the theorem-bearing articulation burden is closed at the declared scope of CF000. The manuscript has established:

- higher-resolution degree character,
- constitutive invariant-bearing of admission,
- higher-resolution admission,
- contextual noncoincidence,
- apartness,
- pre-metric orthogonal rotation,
- same-axis saturation,
- the 0D $\rightarrow$ 1D $\rightarrow$ 2D route,
- the 1D axis-proper effective invariant,
- continuous same-axis bifurcation,
- 1D bearing of the invariant twice,
- seriality / proto-natural-number structure at 1D,
- inherited invariant stacking at 2D,
- the 2D axis-proper effective invariant,
- first numerical relation-grid structure at 2D,

- algebra unlock,
- recursive restatement.

No carrier, metric, QGT, or differentiability theorem has been introduced. Nothing in Section 4 licenses those later objects retroactively. Their absence here is not a gap; it is the stage boundary.

5. Derived Physical Consequences

5.1 A8 visibility note

Forward note for CF00. The same logic that forces orthogonal re-articulation when a domain saturates also makes the later A8 program visible in embryo. The present document does not prove the quantitative logarithmic law. What it proves is the structural source of it:

- one invariant,
- finite lawful expression within a currently admitted domain,
- saturation without discharge,
- orthogonal re-articulation into a new domain. This is why hierarchy becomes visible here without becoming primitive here.

5.2 Forward note on effective invariants

Derived structural consequence with forward note for CF00.

There is only one universal invariant at origin. New domains do not introduce independent primitive invariants. Rather, each admitted axis/domain inherits the origin invariant together with all previously realized effective invariants and adds a new effective invariant proper to that admitted axis/domain. Thus the invariant structure is cumulative and nested: primitive at origin, effective and inherited across successive articulations.

At the scope of the present closure, this rule is already theorem-bearing for the first two admissions. At 1D, the admitted axis/domain immediately acquires the effective invariant of continuous same-axis bifurcative articulation. At 2D, the prior stack is inherited and a new second-axis effective invariant is added. What remains deferred to CF00+ is not the existence of axis-proper effective invariants at 1D and 2D, but the later identification of richer mature forms carried by subsequent domains and later physical realizations.

5.3 A forward empirical question

Forward note for later CFs. If later black-hole physics is read as a domain-relative saturation event, then the horizon/singularity complex should preserve the same structural pattern: same-domain articulation exhausts, the invariant is not lost, and a new lawful channel carries the continuation. That later branch can be tested by asking whether invariant-bearing structure is inherited across the proposed boundary transition.

5.4 Interpretive extensions

Terms such as “tachyonic instability,” “spinodal,” or later field-theoretic realization may be useful bridges to later papers. They are not needed for the theorem-bearing spine of CF000, so they are treated here as forward vocabulary rather than as primitive ingredients.

6. Structural Validation by Finite Witnesses

6.1 Exact finite checks

The theorem-bearing closure of CF000 is carried in Sections 3 and 4. This section adds hard-checkable finite witnesses for the same structure. The witnesses are not empirical evidence from a later runtime and they do not replace the proofs. They are exact finite-cardinality checks: every reported value is obtained by Boolean or integer enumeration under the definitions in Section 2.6.

Table 1: Structural witness values reported in this paper. All tolerances are exact because the checks use Boolean and integer arithmetic.

Witness	Reported value	Method	Tolerance
Primitive survivor count	1 survivor	Exhaustive Boolean enumeration of the candidate classes in Table 2.	0
Same-axis saturation index	9	Finite decision witness $S(n) = \min(n, 9)$; the first attempted same-axis continuation beyond the boundary carries no new same-axis degree.	0
Two-axis placements	16	Exhaustive grid $G_2 = \{0, 1, 2, 3\}^2$.	0
Nonzero two-axis relation vectors	15	$G_2 \setminus \{(0, 0)\}$.	0
Cross-axis-bearing relation vectors	12	Relation vectors (a, b) with $b > 0$ in $G_2 \setminus \{(0, 0)\}$.	0
Pre-2D cross-axis-bearing relation vectors	0	Single-axis restriction $G_1 = \{(a, 0) : a \in \{0, 1, 2, 3\}\}$.	0

6.2 Primitive-candidate enumeration witness

The first finite check asks which primitive candidate can satisfy the admissibility test

$$\text{Adm}(x) = \text{NV}(x) \wedge \neg \text{Flat}(x) \wedge \text{BearOpp}(x) \wedge \text{OneOrigin}(x) \wedge \text{NoDischarge}(x).$$

The result is exact: among the listed primitive candidate classes, only the invariant-bearing origin witness ω satisfies every column.

Table 2: Primitive-candidate enumeration witness. The columns are Boolean values for the finite admissibility language in Section 2.6. The exact survivor count is 1.

Candidate	NV	$\neg \text{Flat}$	BearOpp	OneOrigin	NoDischarge	Adm
isolated 0	0	0	0	1	0	0
isolated 1	1	0	0	1	0	0
third terminal, empty	0	0	0	1	0	0
third terminal, full-flat	1	0	0	1	0	0
external split (0, 1)	1	1	1	0	0	0
internal witness ω	1	1	1	1	1	1

This table is the finite counterpart of Theorems 3.2.1–3.6.1. The two isolated poles fail in different ways, every attempted third terminal collapses into one of those two cases, and the external split fails because CF000 does not allow prior apartness at the primitive layer. The only surviving candidate is one origin-condition bearing the opposition internally.

6.3 Counterexample attempts

The following table records the smallest counterexample attempts a reader might try. Each attempt fails locally, and the failing condition is explicit.

6.4 Saturation-band decision witness

The same-axis saturation witness uses the finite decision rule

$$S(n) = \min(n, 9).$$

This is a compact finite model of the Section 4.7 logic: same-axis articulation can add genuinely new same-axis degrees up to the boundary, but an attempted next genuinely new burden cannot be counted as more of the same saturated axis. It must either duplicate what is already borne or require a new irreducible axis.

The numerical prediction of this finite witness is therefore:

$$S(9) = 9, \quad S(10) - S(9) = 0, \quad \text{second-axis trigger at } n = 10.$$

The companion notebook reproduces this table and exposes the same calculation as executable SymPy arithmetic.

Table 3: Minimal counterexample attempts and the local reason each one is excluded.

Attempt	Local failure	Relevant result
A third terminal pole with no realized content	It is complete realized absence and collapses to 0 .	Theorem 3.5.2
A third terminal pole with realized content but no internal non-coincidence	It is undifferentiated totality and collapses to 1 .	Theorem 3.5.2
A primitive origin made from two already separated poles	It violates the no-prior-apartness discipline; the origin is no longer one primitive condition.	Section 2.4
An admitted articulation that stops bearing the invariant	Non-bearing is discharge; discharge collapses toward an excluded terminal pole.	Theorem 4.3.1
A saturated 1D axis with further same-axis novelty	If the novelty is genuinely same-axis, the axis was not saturated; if it is not genuinely new, it is renaming.	Theorem 4.7.3
A 2D stage without a second irreducible axis	It is not 2D under the stage definition; it remains a one-axis representation.	Definition 4.8.1

Table 4: Saturation-band witness. The first attempted continuation beyond index 9 has no new same-axis degree; it is the finite witness for orthogonal re-articulation rather than same-axis continuation.

Attempt n	$S(n)$	same-axis increment	Decision
0	0	–	seed state before same-axis accumulation
1	1	1	first same-axis articulation
2–8	2–8	1 per step	lawful same-axis continuation
9	9	1	saturation boundary reached
10	9	0	no new same-axis degree; genuinely new continuation requires second axis

6.5 Relation-grid exhaustive scan

The 2D unlock is checked by the finite grid

$$G_2 = \{0, 1, 2, 3\} \times \{0, 1, 2, 3\}.$$

The grid has 16 placements. Removing the origin leaves 15 nonzero relation vectors:

$$R_2 = G_2 \setminus \{(0, 0)\}, \quad |R_2| = 15.$$

A single-axis restriction has only

$$G_1 = \{(0, 0), (1, 0), (2, 0), (3, 0)\},$$

so it has no relation vector with nonzero second-axis component. The second axis therefore does not merely rename one-axis seriality; it makes cross-axis relation possible.

Table 5: Relation-grid counts. The two-axis grid yields 15 nonzero relation vectors from the origin, of which 12 require the second coordinate. The single-axis restriction yields no cross-axis-bearing relation vectors.

Structure	Placements	Nonzero relation vectors	Cross-axis-bearing vectors
single-axis restriction G_1	4	3	0
two-axis grid G_2	16	15	12

For transparency, the 15 nonzero two-axis relation vectors are

$$\begin{aligned}
&(0, 1), (0, 2), (0, 3), \\
&(1, 0), (1, 1), (1, 2), (1, 3), \\
&(2, 0), (2, 1), (2, 2), (2, 3), \\
&(3, 0), (3, 1), (3, 2), (3, 3).
\end{aligned}$$

The values in Table 5 are therefore directly checkable:

$$|G_2| = 4 \cdot 4 = 16, \quad |R_2| = 16 - 1 = 15, \quad |\{(a, b) \in R_2 : b > 0\}| = 4 \cdot 3 = 12.$$

7. Worked Example or Minimal Realization

7.1 Minimal realization setup

Use the weakest stage-correct realization compatible with the theorem spine:

- two terminal primitive poles 0 and 1,
- one realized origin witness ω with $\text{Orig}(\omega)$, $\text{Real}(\omega)$, and $\text{Inv}(\omega)$,
- one first admitted axis/domain $A1$ supporting same-domain differentiated determinations,
- a same-axis saturation condition $\text{Sat}(A1)$ meaning no genuinely new lawful invariant-bearing articulation remains available within $A1$, so further purported articulation there would be either mere renaming or would require a new irreducible axis/domain. No metric, carrier, graph, Laplacian, smooth support, or potential ansatz is assumed in this worked realization. It is a logical realization of the theorem spine only.

7.2 Expected behavior

The worked realization should exhibit: 1. exclusion of stable realized branch-collapse into either isolated pole, 2. non-flat articulation from invariant-bearing origin, 3. higher-resolution determination within the initial axis/domain, 4. same-axis saturation in the strict sense that no genuinely new lawful same-axis articulation remains available, without discharge, 5. need for an irreducible second axis/domain for further lawful articulation, 6. ordered-pair/algebraic description only after that second axis/domain is admitted.

7.3 What the example should show

A successful minimal realization makes the following structure visible:

- no third terminal primitive pole is needed,
- 1D carries an axis-proper effective invariant rather than functioning as a neutral container,
- continuous same-axis bifurcation licenses seriality, repeatable more/less, and proto-natural-number structure at 1D,
- same-axis saturation means exhaustion of lawful same-axis articulation rather than discharge,
- continued genuinely new lawful articulation in the exhausted axis/domain requires a second irreducible axis/domain,
- 2D inherits the prior invariant stack and immediately acquires its own axis-proper effective invariant,
- a first numerical relation-grid becomes possible only once that second irreducible axis/domain is admitted,
- algebraic ordered-pair description becomes meaningful only after the second irreducible axis/domain is admitted.

The same example also gives useful boundary checks. A future realization would extend or revise this formalism if it produced a third terminal primitive survivor, a 1D domain with no axis-proper effective invariant, same-axis saturation with continued same-axis novelty, loss of inherited invariant structure at 2D admission, or algebraic structure before the second irreducible axis appears.

7.4 Algebraic witnesses of the invariant spine

The following three cases demonstrate that the invariant logic proved in Sections 3–4 is generative, not merely classificatory. Each case shows a familiar mathematical object arising directly from the CF000 spine without importing external mathematical machinery. These are not proofs of those objects; they are recognition claims showing the spine already contains their structural source.

7.4.1 The imaginary unit i as ORS expressed algebraically

Status: Derived structural consequence. Scope-correct for CF000.

The 2D algebra unlock (Theorem 4.9.1) licenses ordered pairs (a, b) relative to two irreducible axes. Within this structure, consider the question: what is the algebraic operator whose repeated application cycles through the two-axis structure?

By Theorem 4.8.8, the two admitted axes are irreducible from each other and carry the inherited invariant opposition between them. An operator R acting on ordered pairs that rotates the pair by one axis-step must satisfy:

- $R(1, 0) = (0, 1)$ (maps the first axis to the second),
- $R(0, 1) = (-1, 0)$ (maps the second axis back, inverted, because the two poles are irreconcilable and returning along the same path reaches the opposite pole),
- therefore $R^2(1, 0) = (-1, 0)$, i.e. $R^2 = -1$ on the first-axis unit.

This is exactly the defining property of the imaginary unit: $i^2 = -1$. The operator R is i . Its existence is not an external postulate; it is the first algebraic expression of ORS (Theorem 4.6.1) on the 2D ordered-pair structure. The reason $i^2 = -1$ rather than $+1$ is that returning along the orthogonal rotation path reaches the opposite pole of the invariant, not the same one; the two-pole irreconcilability of the primitive invariant is precisely what prevents $R^2 = +1$. Complex arithmetic is therefore the natural algebra of a two-axis invariant-bearing structure — not an assumption imported into it.

Corollary: first algebraic expression of the primitive tension. The primitive 0/1 tension, after 2D unlock, finds its first orthogonal algebraic expression in an operator R satisfying $R^2 = -I$. This operator is the complex unit i . The point is not that i is assumed at the origin; it is that i becomes the first algebraic witness once the second irreducible axis has been earned.

7.4.2 $0! = 1$ as the identity element of proto- $\mathbb{N}$

Status: Derived structural consequence. Scope-correct for CF000.

Corollary 4.8.7 licenses proto-natural-number structure at 1D: a non-terminating sequence of successor steps generated by the two-pole bifurcation of the invariant. Within this structure, the factorial $n!$ counts ordered arrangements of n successor-elements. For $n = 0$, there are no elements to arrange. The number of ways to arrange nothing is exactly one: the empty arrangement.

This is not a convention. The empty arrangement corresponds to the identity element of the binary combination licensed by Theorem 4.9.1. Specifically, the 2D algebra unlock licenses lawful binary combination; every such combination structure requires an identity element under which combination returns the original. The identity element under ordered arrangement is the empty arrangement, which is unique. The count of unique empty arrangements is therefore 1.

Corollary: empty arrangement identity. Within the proto-natural-number structure licensed at 1D and the binary combination structure licensed at 2D, the empty arrangement is the unique identity for ordered arrangement. Therefore the arrangement count at zero successors is exactly

$$0! = 1.$$

Proof sketch. A combination law without an identity cannot preserve a prior arrangement under combination with no new successor. The no-successor case has exactly one such neutral arrangement: the empty arrangement. Hence the count is 1.

The pole mark **1** (absolute undifferentiated totality, excluded as terminal but preserved as the identity of combination) is the structural ancestor of this result.

7.4.3 Non-scope note: the Pythagorean theorem

Status: Forward note. Out of scope for CF000.

The Pythagorean theorem $a^2 + b^2 = c^2$ requires a metric on the carrier domain, which in turn requires the carrier construction and the induced metric sector derived from the QGT. Both belong to CF00 and CF01 respectively. The theorem is therefore not within CF000's scope boundary and is not demonstrated here. It is listed only to mark the stage boundary explicitly: the metric structure it requires is downstream of the handoff carried in Section 9.5.

8. Companion Notebook and Executable Traceability

The included notebook, *CFN000_Primitive-Origin.ipynb*, is a numerical companion for exploration and reproducibility. It mirrors the main paper at the level appropriate for a notebook: finite witness models, small plots, measured quantities, and a compact JSON-style summary of what was examined.

8.1 Notebook traceability map

Paper section	Notebook section	Quantities instantiated	Reader-facing output
Sections 3–4	Primitive candidate enumeration	terminal poles, third-pole attempts, witness ω	exact survivor table and survivor count 1
Section 4	Articulation chain	$\text{Deg}_{\min}^{\text{g}}$, higher-resolution witnesses	0D–1D–2D stage map
Section 4.7 and Section 6.4	Saturation decision witness	$S(n) = \min(n, 9)$, saturation index 9, trigger at 10	exact saturation table and curve
Sections 4.8–4.9 and Section 6.5	Relation-grid scan	G_1, G_2 , 15 nonzero 2D relation vectors	grid visualization and exact counts
Section 6.3	Counterexample attempts	third-pole, non-bearing, saturated-axis, and pre-2D algebra attempts	negative-control table

The notebook is intentionally supportive. It helps readers inspect the structure, rerun the finite witnesses, and extend the companion layer without interrupting the main argument of the paper.

9. Assumptions, Scope, and Open Questions

9.1 Assumptions used in the derivation

- the branch under study is realized,
- the primitive layer admits only the two terminal poles as terminal primitive extremes,
- no prior apartness or external two-thing reading is available at origin,
- terminal discharge into either isolated pole is excluded for the realized branch,
- later-stage machinery is deferred rather than used as primitive input.

9.2 Scope of applicability

This CF closes the branch-level route 0D–1D–2D. It prepares the handoff to carrier construction, QGT, metriplectic field dynamics, and differentiability, but it does not attempt to derive those later structures here.

9.3 Claims established here

- forced admissible origin,
- incompatibility of invariant-bearing with flatness,
- differentiated articulation and $\text{Deg}_{\min}$,
- constitutive invariant-bearing of admitted articulation,
- HRDCL,
- HRDAP,
- contextual noncoincidence and apartness,
- pre-metric orthogonal rotation,
- same-axis saturation,
- 0D $\rightarrow$ 1D $\rightarrow$ 2D closure,
- the 1D axis-proper effective invariant,
- continuous same-axis bifurcation,
- 1D bearing of the invariant twice,
- seriality / proto-natural-number structure at 1D,
- inherited invariant stacking at 2D,

- the 2D axis-proper effective invariant,
- first numerical relation-grid structure at 2D,
- algebra unlock,
- recursive restatement.

9.4 Claims left for later work

- carrier-family construction,
- QGT,
- named J, M, or $J \oplus M$ split,
- metric or Laplacian derivation,
- differentiability,
- exact A8 constants and interface fractions,
- the full later tower of effective invariants beyond the 1D and 2D instances closed here,
- tachyonic field realization.

9.5 Open questions and later handoff

The next paper, CF00, should begin from the inherited post-2D invariant stack proved here rather than from a primitively assumed carrier. Its natural burden is to derive the first lawful host/carrier consequences of this closure. The quantitative A8 program, the fuller tower of effective invariants beyond the 1D and 2D cases, and any physical field-theoretic realization all belong downstream of the CF000 handoff.

10. Conclusion

CF000 has established that the admissible primitive origin for the realized branch is one condition bearing unresolved 0/1 opposition internally. From that origin, the paper derived non-flat distinguishability, the first admitted axis of articulation, continuous same-axis bifurcation, same-axis saturation, forced orthogonal re-articulation into a second irreducible axis, and the first numerical relation-grid in which algebra becomes meaningful.

The structural witnesses in Section 6 make the derivation directly checkable in finite form: exactly one primitive candidate survives the admissibility test, the same-axis decision witness saturates at index 9, the next genuinely new continuation triggers a second-axis requirement, and the first two-axis grid contains 15 nonzero relation vectors. These finite checks do not replace the proofs; they make the root spine easy to inspect.

The handoff is now precise. CF000 closes the pre-carrier 0D–1D–2D route and passes the inherited post-2D invariant stack to CF00, where the first lawful carrier consequences can be derived.

11. Integration with Broader VDM Theory

CF000 is the root branch in the present dependency order. Its completed handoff is the post-2D invariant stack: primitive unresolved origin, first articulation, continuous same-axis bifurcation, saturation, forced orthogonal re-articulation, numerical relation-grid structure, and algebra unlock.

The next papers should be read as continuations of that handoff. CF00 begins from the inherited post-2D stack and derives the first lawful host/carrier consequences. CF01 then develops the QGT and the later geometric/metriplectic machinery. In this order, later structures enrich the root formalism without being used to prove it retroactively.

This makes the integration rule straightforward: CF000 supplies the root dependency spine; CF00 supplies the first carrier/geometry stage; CF01 and later papers supply progressively richer physical descendants. Each stage inherits the previous burden and adds its own effective invariant structure.

12. References and Provenance

The main derivation is self-contained in Sections 3 and 4. The materials listed here provide provenance, context, and downstream alignment for readers who want to follow the wider VDM development.

A(-1) is cited as an upstream statement of the orthogonality-invariant law and continuation trigger. In the dependency order of this manuscript, the primitive origin is the unresolved invariant tension of **0** and **1** borne as one primitive condition. The symbol i is treated here as the first orthogonal/algebraic expression of that tension once two-axis articulation is licensed. H000 and related origin documents preserve proposal-grade and branch-resolution context, while CF00 and CF01 are downstream alignment references.

12.1 Internal constitutional / provenance and downstream alignment references

References

Internal Constitutional and Provenance References

- [1] J. K. Lietz, “Universal Orthogonality Invariant: The Generative Nature of i ” (A(-1), v0.3), *Zenodo*, 2026. doi: 10.5281/zenodo.19211725. Cited for the upstream orthogonality-invariant law, the non-discharge discipline, and the universal continuation trigger. In the dependency order used here, i is read as the first orthogonal/algebraic expression of the primitive **0/1** tension after two-axis articulation is licensed.
- [2] J. K. Lietz, “H000: Primitive Bifurcation As A Minimal Generative Origin For Distinguishability, Hierarchy, And Carrier Emergence” (v0.1), *Zenodo*, 2026. doi: 10.5281/zenodo.19009927. Cited for proposal-grade provenance of the primitive bifurcation hypothesis and for continuity with the earlier research branch.
- [3] J. K. Lietz, “Branch Relative Origin Resolution” (v0.1), *Zenodo*, 2026. doi: 10.5281/zenodo.19010844. Cited for branch-relative origin-resolution provenance and terminology alignment.
- [4] J. K. Lietz, “The Lietz Infinity Resolution Conjecture: Logarithmic Hierarchical Interfaces as the Tachyonic Pulled-Front Re-Expression of the Primitive Invariant” (v1.1), *Zenodo*, 2026. doi: 10.5281/zenodo.19155667. Cited for downstream A8/LIR provenance and later-regime re-expression of the primitive invariant.

Downstream Canon Alignment References

- [5] J. K. Lietz, “CF00: Induced Geometry and Emergent Dynamics,” *Zenodo*, 2026. doi: 10.5281/zenodo.19380518. Cited for downstream alignment. CF00 inherits the post-2D invariant stack closed here and begins the carrier/geometry stage.
- [6] J. K. Lietz, “CF01: Quantum Geometric Tensor Decomposition to Metriplectic Brackets,” *Zenodo*, 2026. doi: 10.5281/zenodo.19380435. Cited for downstream alignment. QGT, metric curvature, and metriplectic brackets enter at the later geometric stage.

12.2 Structural and executable companion artifacts

The following items support structural witness checking, executable traceability, and later witness development.

- *CFN000_Primitive_Origin.ipynb*, paired notebook artifact for primitive enumeration, saturation, relation-grid counting, counterexample attempts, and SymPy-backed exact arithmetic.
- *lean/CF000_Structural_Witness.lean*, Lean 4 surface file for the finite Boolean and integer witness claims.
- *scripts/cf000_structural_witness_sympy.py*, SymPy script that regenerates the structural witness JSON summary.
- *outputs/cf000_structural_witness_summary.json* and *outputs/cf000_structural_witness_sympy.json*, exact finite-witness summaries.

12.3 Local symbol and closure summary

Symbol	Meaning	Status
0	Absolute nullity	Primitive
1	Absolute undifferentiated totality	Primitive
$\mathcal{C}$	Logical domain of candidate realization conditions	Primitive setup
$\text{Orig}(x)$	Origin-candidate predicate	Primitive
$\text{Ster}(x)$	Sterility predicate	Primitive
$\text{Flat}(x)$	Flatness predicate	Primitive
$\text{NV}(x)$	Non-vacuousness predicate	Primitive

$\text{Real}(x)$	Realizability predicate	Primitive
$\text{Opp}(x, y)$	Opposition predicate	Primitive
$\text{Inv}(x)$	Primitive invariant carriage shorthand: unresolved 0/1 pole-opposition borne internally without terminal discharge	Derived shorthand
$\text{Diff}(x)$	Differentiated articulation	Derived
$\text{Deg}_{\min}(x)$	Lowest-resolution invariant-bearing differentiated degree	Derived
ω	Realized branch-origin witness	Witness
i	First orthogonal/algebraic expression of the undischarged primitive tension after two-axis articulation is licensed; not the CF000 origin itself	Derived algebraic witness
1D axis-proper effective invariant	Continuous same-axis bifurcative articulation	Derived
Continuous same-axis bifurcation	Indefinitely reiterable polar articulation along 1D	Derived
Serial / proto- $\mathbb{N}$ structure	Successor structure from non-terminating 1D bifurcation	Derived structural consequence
Inherited invariant stacking	Preservation of prior invariant stack at each new axis admission	Derived
Numerical relation-grid invariant	2D axis-proper effective invariant of inter-axis irreducible numerical relation	Derived
First numerical relation-grid	Cross-axis determination structure first available at 2D	Derived structural consequence
Algebra unlock	First licensing of ordered-pair description and lawful binary combination at 2D	Derived

Primitive Root Package

- nullity and undifferentiated-totality exclusion
- no third terminal primitive pole
- forced admissible origin
- invariant-bearing incompatible with flatness
- differentiated articulation / $\text{Deg}_{\min}$
- constitutive invariant-bearing of admission
- HRDCL
- HRDAP
- contextual noncoincidence / apartness
- 1D admission
- 1D axis-proper effective invariant
- continuous same-axis bifurcation
- 1D bearing of the invariant twice
- serial / proto-natural-number structure
- ORS
- same-domain saturation
- 2D admission by forced orthogonal re-articulation
- inherited invariant stacking at 2D
- 2D axis-proper effective invariant (numerical relation-grid invariant)
- first numerical relation-grid structure
- algebra unlock
- i as ORS expressed algebraically after algebra unlock, not as a replacement for the primitive 0/1 tension-origin

→ recursive restatement

The dependency order remains forward: the root derivation is carried first, and later empirical notes, notebook witnesses, and bibliographic links help readers continue the work without changing the proof order.